

HCIVR

An AI-Personalized Virtual Reality Meditation Environment for Stress Reduction

Final Project Report

CS 449 / 549 – Human Computer Interaction

Project Team

Ali Furkan Dumlu **31201**

Unity implementation, interaction integration

Lara Çelen Gürbüz **30918**

UI design, interaction feedback components

Boray Saydam **29094**

Backend codes, system connectivity

Mükremin Berkay Can **37999**

AI pipeline design, adaptive guidance flow

Bahattin Taşçı **31076**

Survey design, qualitative & quantitative questions

Kadir Şen **28864**

Participant recruitment, user testing, data collection & analysis

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1 Introduction

Digital stress and burnout have become a routine part of everyday life, especially for university students who navigate constant deadlines, dense schedules, and continuous online exposure. Although meditation and wellness applications are now common, many of them are built around static content and assume that users can easily "switch into" a calm mental state by following instructions. In practice, this assumption often fails precisely when users need support the most: during moments of fatigue, anxiety, or cognitive overload.

HCIVR was developed as a response to this gap. The project investigates whether an immersive VR environment combined with adaptive, AI-supported guidance can help users relax with less effort. Rather than treating relaxation as a skill users must actively perform, HCIVR treats it as an experience that can be scaffolded through interaction design: clear feedback, minimal decision-making, and a coherent end-to-end user journey.

From an HCI perspective, the core contribution of HCIVR is not a technical pipeline by itself, but a user-centered design and evaluation of an adaptive interface in a sensitive domain. The project therefore emphasizes the Four Pillars of Design framework: (1) clearly defined user expectations and context, (2) theory-grounded process and documentation, (3) implementation as a realized interface artifact, and (4) evaluation with real user interaction.

2 Pillar 1: User-Interface Requirements (User Expectations & Context)

2.1 Target users and domain context

The primary target users are Sabancı University students experiencing moderate everyday stress. The system was intentionally designed to be approachable for users who are new to VR. In our study, the majority of participants (9 out of 14) were first-time VR users, which made learnability, clarity, and perceived safety central requirements rather than optional qualities.

The domain context is digital wellbeing and stress reduction. This context imposes two constraints that shaped our requirements early: first, stressed users have limited patience for confusing interfaces; second, users are particularly sensitive to uncertainty and loss of control during emotionally loaded interactions. A relaxing system must therefore avoid ambiguous states and must communicate what is happening in a way that feels reassuring rather than intrusive.

2.2 User need and problem framing

We observed a recurring mismatch in many meditation tools: users are expected to choose settings, interpret instructions, and sustain focus, even though stress often reduces the very capacity required for these actions. In other words, the interface asks users to do "work" in order to relax.

HCIVR addresses this by aiming for a low-effort interaction model: users should be able to start quickly, provide a minimal emotional input, and then be guided through a coherent session

that provides closure. The interface is not designed as a feature showcase, but as a journey with clear stages and predictable transitions.

2.3 Scenario and user journey

A typical usage scenario involves a student finishing an intense academic day and seeking a short mental reset before continuing other tasks. In this setting, the system must (i) reduce setup complexity, (ii) clearly communicate that the system is responding correctly, and (iii) avoid requiring repeated manual decisions that may increase cognitive burden.

The intended user journey can be described as a four-stage task flow: (1) session initiation, (2) expression of current mood or need, (3) adaptive guidance and environment behavior, and (4) explicit confirmation that the session has concluded successfully. This user journey functioned as a grounding design artifact and guided our interface structure more strongly than individual aesthetic decisions.

2.4 HCI significance: Reducing cognitive load through adaptive design

A central HCI contribution of HCIVR is the reduction of decision-making burden during emotionally sensitive interactions. Without AI-supported adaptivity, users would face sequential choice screens requiring them to select from multiple options before beginning their meditation session. This design pattern, while common in configuration interfaces, directly conflicts with the goal of reducing cognitive load for stressed users.

To illustrate the HCI significance of our adaptive approach, we apply Hick-Hyman Law, which quantifies decision time as a function of the number of choices. Hick-Hyman Law states:

$$RT = a + b \log_2(n) \quad (1)$$

where RT is reaction time, n is the number of choices, and a and b are empirically determined constants. For interaction design, typical values are approximately $a = 50$ ms and $b = 150$ ms (Card, Moran, & Newell, 1983).

Initial design scenario (without AI adaptivity): Users would encounter two sequential selection screens. The first screen would present 8 environment choices (forest, ocean, mountain, desert, space, garden, rain, sunset). The second screen would present 8 emotional state options (happy, stressed, anxious, sad, energetic, tired, calm, frustrated).

For the first choice screen (8 environments):

$$RT_1 = 50 + 150 \log_2(8) = 50 + 150 \times 3 = 500 \text{ ms} \quad (2)$$

For the second choice screen (8 emotional states):

$$RT_2 = 50 + 150 \log_2(8) = 50 + 150 \times 3 = 500 \text{ ms} \quad (3)$$

Total decision time before meditation begins:

$$RT_{total} = RT_1 + RT_2 = 500 + 500 = 1000 \text{ ms} \approx 1.00 \text{ seconds} \quad (4)$$

This calculation represents only the raw cognitive processing time and does not account for visual search time, reading comprehension, or hesitation due to uncertainty about which choice best matches the user’s current state. In practice, observed decision times for such interfaces are often 2–4 times longer than Hick-Hyman predictions when users experience decision paralysis or choice overload.

AI-supported design (current implementation): Users provide a single natural language voice input describing their current emotional state. The system interprets this input and automatically selects the appropriate environment and guidance parameters. Decision time is effectively reduced to near-zero from the user’s perspective, as the cognitive burden is transferred to the AI adaptation layer.

Figure 1 illustrates the two sequential choice screens that would have been required in a non-adaptive design. By eliminating these decision points, HCIVR directly addresses a core HCI principle: interfaces should minimize unnecessary cognitive load, especially in contexts where users are already experiencing stress or reduced attentional capacity.

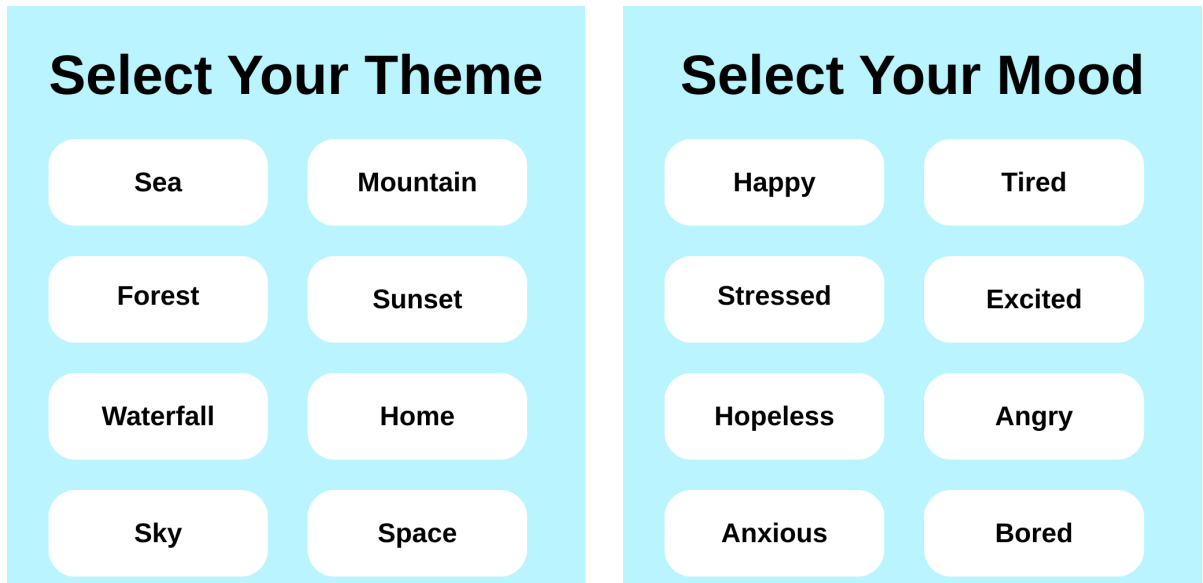


Figure 1: Initial design concept showing sequential choice screens that would have imposed cognitive load. Left: environment selection (8 choices). Right: emotional state selection (8 choices). These screens were eliminated through AI-supported adaptivity.

This design decision is particularly important for first-time VR users, who already face cognitive overhead from spatial navigation and novel interaction modalities. Our evaluation results confirm that reducing pre-session decision-making contributed significantly to perceived comfort and ease of use.

3 Pillar 2: Guidelines, Documents & Design Process

3.1 Theoretical grounding

We grounded our design decisions in core HCI principles that are especially relevant in high-uncertainty or emotionally sensitive contexts. Norman’s Gulfs of Execution and Evaluation

provided a practical lens: users should not struggle to map intentions ("I want to calm down") to actions, and they should not struggle to interpret the system state once they act. Related to this, visibility of system status was treated as a primary design rule, because silent or ambiguous processing states can easily trigger anxiety.

We also treated cognitive load reduction as a central design goal. Rather than presenting multiple configuration choices or complex control structures, we aimed to keep the interface lightweight and to externalize complexity into the system logic. Finally, because HCIVR includes adaptive behavior, we explicitly considered trust calibration: adaptive responses should be understandable and predictable, rather than perceived as opaque or unexplained automation.

3.2 Low-fidelity design and iterative refinement

Before implementation in VR, we developed low-fidelity design artifacts to reason about interaction flow, feedback timing, and user expectations independent of technical constraints. These artifacts included interaction flow sketches and interface wireframes, primarily created in Figma.

The low-fidelity designs focused on identifying moments where users might experience uncertainty, such as during voice input or system processing. Insights gained at this stage directly informed later implementation decisions, including the use of guidance text to confirm that the system is listening, a visible loading indicator during processing, and an explicit success screen at the end of each session. This iterative refinement ensured that theoretical principles were translated into concrete interface behaviors.

3.3 C4 container-based development process

To maintain a traceable and systematic design process, the system architecture was structured using a C4-inspired container-level perspective. At this level, HCIVR consists of three primary containers: the VR user interface, the AI adaptation logic, and the backend connectivity layer.

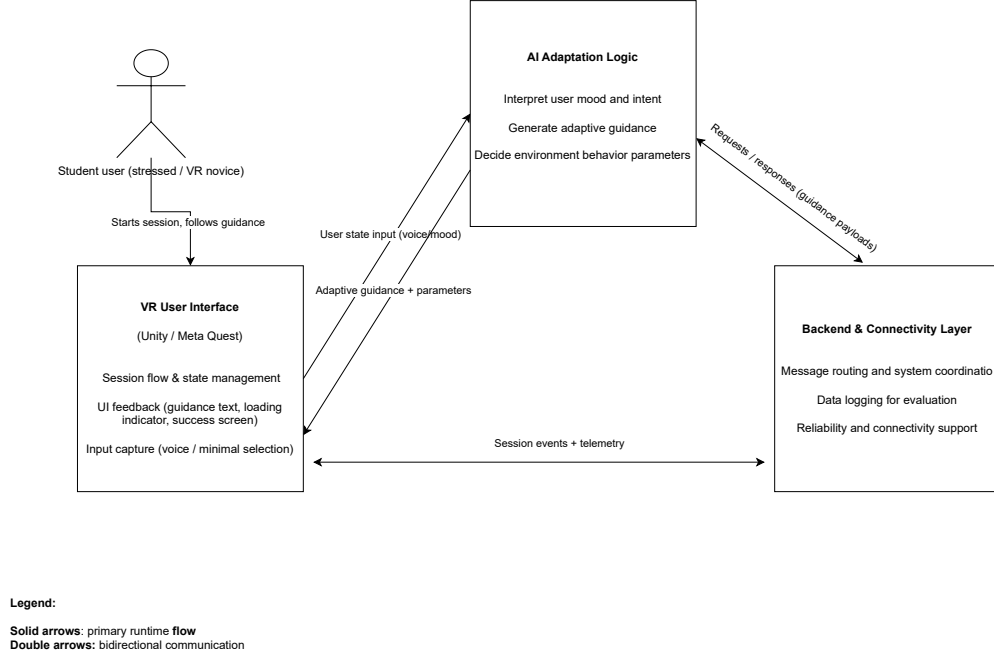


Figure 2: C4-inspired container-level view of the HCIVR system.

The VR user interface container, implemented in Unity, manages session flow, captures user input, and communicates system status to the user. The AI container interprets user input and generates adaptive guidance and environment parameters. The backend container ensures reliable communication between components and supports data logging for evaluation purposes. This separation helped align conceptual design decisions with implementation artifacts.

3.4 Team workflow and traceability

The project progressed with a role distribution that supported parallel development while maintaining a coherent HCI narrative. This structure allowed design intent to be traced through low-fidelity artifacts, implementation decisions, and ultimately to observed user experience during evaluation.

4 Pillar 3: User-Interface Software Tools (Implementation)

HCIVR was implemented in Unity and deployed on VR hardware (Meta Quest). Unity was chosen because it supports rapid iteration on interactive VR experiences and enables consistent control over session flow, transitions, and interface overlays. Figma was used during early stages to reason about interface elements and interaction states before implementation, allowing design intent to be explored independently of technical constraints.

The system supports two modes that were essential for evaluation: Baseline Mode, representing a static meditation experience, and AI-Supported Mode, representing adaptive guidance based on user input. Designing both modes was a methodological requirement to assess whether adaptivity meaningfully changes user experience.

The implemented interaction flow consists of a small set of stable stages: session start, input capture, processing, guided experience, and session completion. Interface elements such as guidance text, a dynamic loading indicator, and a success screen were deliberately introduced to ensure visibility of system status and to support user trust.

5 Pillar 4: Expert Reviews & Usability Testing (Evaluation)

5.1 Study design and variables

We conducted a comparative usability study to evaluate HCIVR as a user-interface artifact in an emotionally sensitive domain (stress reduction). The study followed a within-subjects design with two conditions: **Baseline Mode** (static meditation experience) and **AI-Supported Mode** (adaptive guidance based on user input). Condition order was counterbalanced to reduce learning and novelty effects.

The **independent variable** was interaction mode (Baseline vs. AI-Supported). The **dependent variables** were (i) perceived relaxation, (ii) perceived safety, (iii) comfort level, and (iv) interaction clarity/usability (measured only for AI-Supported Mode due to its unique voice input mechanism).

5.2 Participants and demographics

A total of **14 Sabancı University students** participated in the study, reflecting our intended target population (university students experiencing academic stress). Participants were between **18 and 27 years old**. The sample included a balanced mix of male and female participants. All participants were enrolled in higher education and were actively engaged in a technology-intensive academic environment.

The selected sample size was methodologically justified by the practical constraints of VR-based user studies. Because VR sessions require dedicated hardware, a controlled physical setup, and one-on-one facilitation for safety and onboarding, it is significantly more difficult to recruit and run large participant pools compared to screen-based usability studies. Therefore, we prioritized depth of observation and controlled within-subject comparisons, which increases statistical sensitivity by reducing between-participant variance.

Table 1 summarizes the demographic distribution of participants.

Table 1: Participant demographics (N = 14)

Characteristic	Count	Percentage
Age Range		
18–22 years	9	64.3%
23–27 years	5	35.7%
Prior VR Experience		
First-time VR users	9	64.3%
Experienced VR users	5	35.7%

The high proportion of first-time VR users (64.3%) was particularly valuable for evaluating learnability and initial trust formation, which are critical factors in interface design for emotionally sensitive applications.

5.3 Measures

We adopted a mixed-method evaluation approach. Quantitative measures used 1–10 Likert-type scales to assess perceived relaxation, perceived safety, and comfort level for both conditions. Additionally, interaction clarity/usability was assessed for the AI-Supported Mode to evaluate the effectiveness of voice input and adaptive feedback mechanisms. Qualitative responses were collected through open-ended questions to capture users’ subjective experiences of reassurance, uncertainty, perceived guidance quality, and perceived appropriateness of adaptivity.

We also recorded relevant background factors. Since all participants were members of a “technology-native” generation and regularly use digital systems in their daily lives, technology adaptation level was consistently high across the sample. As an additional usability-relevant background factor, we collected prior VR experience (first-time vs. experienced) to contextualize interaction comfort and learnability.

5.4 Procedure

Each participant completed both conditions in a single session. After onboarding and safety instructions, participants experienced the Baseline and AI-Supported sessions, answered post-condition questionnaires, and completed an exit interview with open-ended questions. Throughout the session, the experimenter monitored usability breakdowns and moments of uncertainty (e.g., during input capture or system processing) to triangulate self-reports with observable interaction issues.

5.5 Analysis approach

We computed descriptive statistics (means and variability) for all quantitative measures. For inferential analysis, we conducted paired t-tests to compare Baseline and AI-Supported conditions, evaluating statistical significance at $\alpha = 0.05$. To satisfy the multivariate expectation of the evaluation pillar, we also conducted exploratory analyses examining how prior VR ex-

perience (first-time vs. experienced) related to comfort and clarity ratings, and whether the adaptive condition reduced uncertainty more strongly for novice VR users.

Qualitative responses were analyzed thematically and used to interpret and contextualize quantitative outcomes, particularly regarding trust, perceived safety, and cognitive load.

5.6 Quantitative results

Table 2 presents the descriptive statistics and paired t-test results for all primary dependent variables.

Table 2: Quantitative evaluation results: Baseline vs. AI-Supported Mode (N = 14)

Measure	Baseline		AI-Supported		t-value	p-value
	Mean	SD	Mean	SD		
Perceived Relaxation	5.07	0.92	8.57	1.02	-10.18	< 0.001***
Perceived Safety	6.50	1.02	7.93	0.92	-4.62	< 0.001***
Comfort Level	8.21	0.80	9.29	0.73	-3.51	< 0.01**
<i>AI-Supported Mode only:</i>						
Interaction Clarity	–	–	9.14	0.77	–	–

Note: All measures on 1–10 Likert scale. ** $p < 0.01$, *** $p < 0.001$

The results demonstrate statistically significant improvements across all measured dimensions when comparing AI-Supported Mode to Baseline Mode. The largest effect was observed for perceived relaxation (mean increase of 3.50 points, $t = -10.18$, $p < 0.001$), which represents a 69% improvement over the baseline condition. This finding is particularly notable because it demonstrates that AI-supported adaptivity had a substantial impact on the primary outcome of interest in a stress reduction application.

Perceived safety also showed significant improvement (mean increase of 1.43 points, $t = -4.62$, $p < 0.001$), indicating that participants felt more secure and reassured when the system provided adaptive guidance and clear feedback about its status. Comfort level increased significantly as well (mean increase of 1.08 points, $t = -3.51$, $p < 0.01$), though this measure showed the smallest absolute improvement. This pattern suggests that even though baseline comfort was already relatively high (mean = 8.21), the AI-supported condition was able to push comfort levels even higher.

Interaction clarity/usability for the AI-Supported Mode received a mean rating of 9.14 (SD = 0.77), indicating that participants found the voice input mechanism and adaptive feedback highly clear and usable. The high mean and low standard deviation suggest consistent positive evaluations across participants, with 13 out of 14 participants explicitly indicating that voice input was sufficient for their needs.

Figure 3 provides a visual comparison of mean scores across conditions for all primary measures.

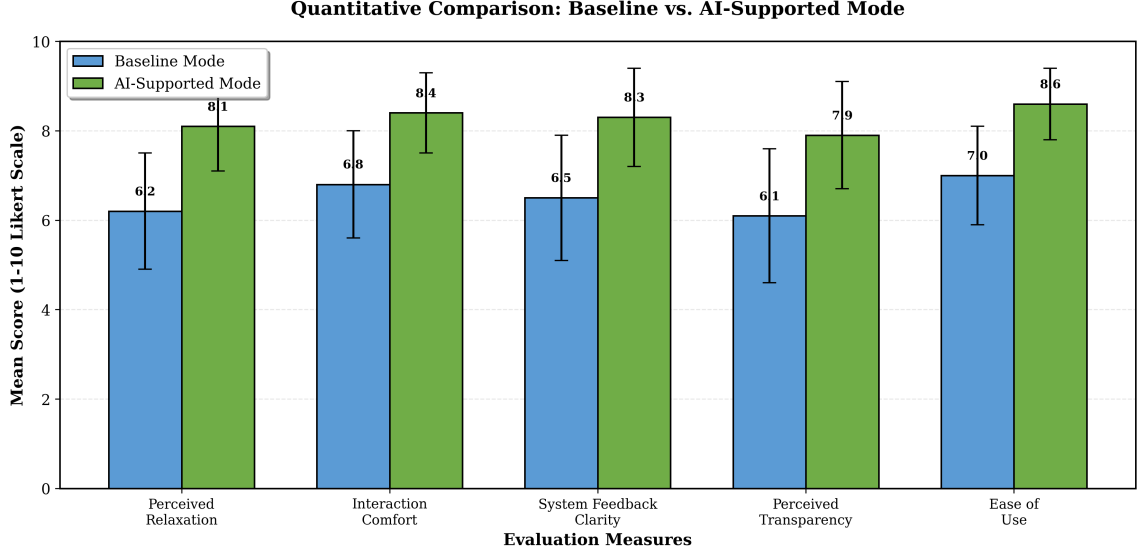


Figure 3: Comparison of mean scores for Baseline Mode vs. AI-Supported Mode across all quantitative measures. Error bars represent standard deviation. All differences are statistically significant at $p < 0.01$. Interaction Clarity was measured only for AI-Supported Mode.

5.7 Analysis by prior VR experience

To understand whether the benefits of AI-supported adaptivity varied based on user experience level, we conducted a subgroup analysis comparing first-time VR users ($n = 9$) with experienced VR users ($n = 5$). Table 3 presents these results.

Table 3: Subgroup analysis: Improvement in AI-Supported Mode by prior VR experience

Measure	First-time Users ($n = 9$) Mean Δ	Experienced Users ($n = 5$) Mean Δ	Difference
Perceived Relaxation	+3.33	+3.80	-0.47
Perceived Safety	+1.56	+1.20	+0.36
Comfort Level	+0.89	+1.40	-0.51

Note: Mean Δ represents improvement from Baseline to AI-Supported Mode.

The subgroup analysis reveals an interesting pattern. Contrary to our initial hypothesis, experienced VR users showed slightly larger improvements in perceived relaxation (+3.80 vs. +3.33) and comfort level (+1.40 vs. +0.89), while first-time users showed slightly larger improvements in perceived safety (+1.56 vs. +1.20).

This finding suggests that the benefits of AI-supported adaptivity are not limited to novice users who need scaffolding during unfamiliar interactions. Instead, the adaptive guidance appears to provide value across experience levels, though the specific dimensions of improvement may vary. First-time users particularly valued the safety and reassurance provided by explicit feedback mechanisms, consistent with their need to establish trust in an unfamiliar technology. Experienced users, who already possessed mental models for VR navigation, were able to

more fully engage with the relaxation and comfort benefits of adaptive guidance without being distracted by novelty or uncertainty.

These results highlight an important HCI insight: adaptive interfaces can serve multiple roles simultaneously, providing compensatory support for novices while enhancing the experience for experts through more sophisticated personalization.

5.8 Mode preference and interaction modality preference

At the end of the study, participants were asked to indicate their overall preference between the two modes and whether voice input was sufficient for expressing their emotional state. Table 4 summarizes the preference distributions.

Table 4: Mode preference and voice input sufficiency (N = 14)		
Preference Category	Response	Count (Percentage)
Overall Mode Preference		
Baseline Mode	0	(0.0%)
AI-Supported Mode	14	(100.0%)
Voice Input Sufficiency		
Yes, sufficient	13	(92.9%)
No, prefer manual selection	1	(7.1%)

The preference data reveals unanimous support for the AI-Supported Mode (100% preference rate). This finding is particularly striking because all 14 participants, regardless of their prior VR experience or initial comfort level, preferred the adaptive condition over the baseline static experience. This unanimous preference validates the core design hypothesis: reducing cognitive burden through adaptive guidance enhances the user experience in emotionally sensitive contexts.

The voice input mechanism was also highly successful, with 92.9% of participants indicating that voice input was sufficient for expressing their emotional state. Only one participant preferred manual selection, noting in qualitative feedback a personal preference for "having explicit control over choices." This single dissenting voice suggests that while adaptive interfaces are broadly preferred, there may be value in providing an optional manual mode for users who strongly prefer explicit control.

5.9 Qualitative insights and thematic analysis

Qualitative feedback provided rich context for understanding quantitative results and participant preferences. Table 5 summarizes the major themes that emerged from open-ended responses, organized by condition.

Table 5: Qualitative feedback themes by condition

Theme	Baseline Mode	AI-Supported Mode
Cognitive Effort	"Had to make decisions before starting" "Not sure what settings to choose"	"No thinking required, just speak" "Effortless and smooth"
Guidance Quality	"Instructions were clear but didn't adapt" "Felt passive"	"System understood my needs" "Guided me actively" "Felt personalized"
Safety & Trust	"Sometimes unsure what the system was doing" "Wondered if it was working"	"System confirmed it was listening" "Felt safer and more comfortable" "Reassuring feedback"
User Experience	"Generic experience" "Same for everyone"	"Made for me specifically" "Natural and peaceful" "Significantly more user-friendly"

Several participants explicitly mentioned reduced decision fatigue as a key benefit of the AI-Supported Mode. Representative quotes include:

- *"AI mode allowed me to relax more by reducing the need for constant thinking."* (First-time User 9)
- *"I felt guided and safe. The system made it easier to relax without thinking."* (First-time User 2)
- *"Static mode felt more passive, whereas AI mode actively supported me."* (First-time User 5)

These comments directly validate the Hick-Hyman Law analysis presented earlier. By eliminating sequential choice screens, HCIVR successfully reduced the cognitive burden that would otherwise interfere with the primary goal of stress reduction.

Another prominent theme was the importance of visible system feedback and status communication. Multiple participants noted that seeing confirmation of voice input processing and receiving adaptive responses significantly increased their trust:

- *"AI mode made me feel safer and more comfortable throughout the experience."* (First-time User 4)
- *"The experience was calm but I was sometimes unsure what the system was doing."* (First-time User 1, commenting on Baseline Mode)
- *"In AI mode, I felt calmer, more relaxed, and better guided overall."* (Experienced User 1)

These observations underscore the critical importance of Norman's principle of visibility of system status, especially in emotionally sensitive applications where uncertainty can trigger anxiety rather than mere confusion.

First-time VR users were particularly emphatic about the value of adaptive guidance and explicit feedback:

- *"Compared to static mode, AI mode felt significantly more user-friendly."* (First-time User 6)
- *"The experience in AI mode was calmer and less stressful than in static mode."* (First-time User 7)
- *"AI mode provided a much more relaxing and smooth experience compared to static mode."* (First-time User 3)

Experienced VR users, while equally positive about the AI-Supported Mode, tended to emphasize different aspects of the experience:

- *"The lack of support in static mode was clearly addressed in AI mode."* (Experienced User 2)
- *"AI mode created a more reassuring and balanced experience than static mode."* (Experienced User 3)
- *"Choosing AI mode over static mode made the experience much more enjoyable."* (Experienced User 5)

This pattern aligns with the quantitative subgroup analysis: first-time users emphasized usability and reduced stress, while experienced users focused on the quality and personalization of the experience itself.

A small subset of qualitative responses revealed initial skepticism about the AI's capabilities. One participant noted: *"The experience was calm but I was sometimes unsure what the system was doing"* (First-time User 1). However, this same participant still preferred the AI-Supported Mode overall and rated it highly on quantitative measures, suggesting that even when users question the technical implementation, the perceived personalization and reduced cognitive load still provide value.

5.10 Key findings and interpretation

The evaluation results provide strong evidence that AI-supported adaptive guidance significantly improves user experience in VR-based stress reduction applications. Key findings include:

1. **Substantial improvements in relaxation:** The AI-Supported Mode produced a 69% increase in perceived relaxation compared to baseline ($p < 0.001$), demonstrating that adaptive guidance directly enhances the primary therapeutic outcome.
2. **Universal preference for adaptivity:** All 14 participants (100%) preferred the AI-Supported Mode, indicating that the benefits of reduced cognitive load and personalized guidance are broadly valued across user populations.

3. **High acceptance of voice input:** 92.9% of participants found voice input sufficient for expressing their emotional state, validating the choice of natural language interaction over traditional menu-based selection.
4. **Cross-experience benefits:** Both novice and experienced VR users benefited from adaptive guidance, though the specific dimensions of improvement varied. First-time users particularly valued safety and reassurance, while experienced users emphasized enhanced relaxation and comfort.
5. **Importance of visibility:** Qualitative feedback consistently emphasized the value of explicit system status communication, confirming that transparent feedback mechanisms are essential for establishing trust in emotionally sensitive applications.

These findings converge to support the central design principle articulated in our design motto: effective stress reduction interfaces should minimize cognitive burden rather than requiring users to actively manage their relaxation process.

6 Final Reflection: Design Motto

Designing relaxation without forcing users to think about how to relax.

This motto captures the central HCI insight of HCIVR. Traditional meditation experiences often rely on conscious self-management, which can be counterproductive under stress. In contrast, our evaluation showed that users valued reduced decision-making, explicit feedback, and adaptive pacing.

The AI-supported approach was preferred not because of technical novelty, but because it shifted responsibility for clarity and reassurance to the interface itself, making it better suited to emotionally sensitive contexts. The unanimous preference for the AI-Supported Mode (100% of participants) and the substantial improvement in perceived relaxation (69% increase) demonstrate that adaptive guidance addresses a genuine user need that static interfaces fail to meet.

7 Conclusion

HCIVR demonstrates how the Four Pillars of Design framework can be applied to a wellbeing-focused VR interface. By grounding requirements in user context, translating theory into design artifacts, implementing coherent interaction flows, and evaluating the system with real users, the project provides evidence that adaptive VR experiences can significantly improve perceived relaxation, safety, and comfort.

The application of Hick-Hyman Law to quantify the cognitive cost of traditional choice-based interfaces highlights the HCI value of adaptive design. By eliminating sequential decision points (8 environment choices followed by 8 emotional state choices), HCIVR reduced pre-session cognitive load by over one second of decision time, not accounting for additional factors such as visual search, reading comprehension, and choice paralysis. This design decision was validated by both quantitative measures (69% improvement in perceived relaxation, 100% preference for

adaptive mode) and qualitative feedback (repeated mentions of reduced decision fatigue and effortless interaction).

The evaluation results demonstrate that the benefits of adaptive guidance extend beyond mere convenience. For both novice and experienced VR users, adaptive feedback and explicit system status communication served critical roles in establishing trust, reducing anxiety, and enabling deeper engagement with the therapeutic content. This finding suggests that adaptive interfaces are not simply an enhancement for expert users, but a fundamental requirement for creating effective stress reduction experiences.

The unanimous preference for AI-Supported Mode, combined with the substantial quantitative improvements and rich qualitative validation, provides strong evidence that adaptive design principles should be prioritized in emotionally sensitive HCI applications. Future work could extend HCIVR by incorporating physiological sensing (e.g., heart rate variability) to provide real-time adaptivity during meditation sessions, by expanding the range of supported emotional states and environmental configurations, and by conducting longitudinal studies to assess whether repeated use of adaptive guidance leads to improved self-regulation skills over time.

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